

Average slope:

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Slope at one point: also known as instant slope: is the derivative of the function

$$\frac{dy}{dx}$$

Practice problems from Prof. Gilbert Strang's Lecture - Big Picture Derivatives

1. For $y = 2x^3$, what is the average slope $= \frac{\Delta y}{\Delta x}$ from $x = 1$ to $x = 2$?
2. What is the instant slope of $y = 2x^3$ at $x = 1$?
3. $y = x^n$ has $\frac{dy}{dx} = nx^{n-1}$. What is $\frac{dy}{dx}$ when $y(x) = \frac{1}{x} = x^{-1}$?
4. For $y = x^{-1}$, what is the average slope $\frac{\Delta y}{\Delta x}$ from $x = \frac{1}{2}$ to $x = 1$?
5. What is the instant slope of $y = x^{-1}$ at $x = \frac{1}{2}$?

Q1,2

$$y = 2x^3$$

given : two points: at $x = 1$ to $x = 2$

we first compute y values at the given points.

$$\begin{aligned} f(1) &= 2(1)^3 = 2 \\ f(2) &= 2(2)^3 = 16 \end{aligned}$$

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{16 - 2}{2 - 1} = \frac{14}{1}$$

then to find the instant slope , we differentiate the given equation.

$$\begin{aligned} f(x) &= 2x^3 \\ d/dx(x) &= 6x^2 \\ \text{at } x = 1, f'(x) &= 6(1)^3 = 6 \end{aligned}$$

q3.

$$\begin{aligned} f(x) &= \frac{1}{x} \\ f(x) &= 1 \cdot x^{-1} \\ f(x)' &= -1x^{(-1-1)} = -1x^{-2} \end{aligned}$$

Q4 &. Q5

for

$$y = x^{-1}$$

what is the average slope from $x = 1/2$ to $x = 1$?

y values at the two given points are:

at $x = 1/2$

$$y = \left(\frac{1}{2}\right)^{-1} = \left(\frac{-1}{2}\right)^{-2} = 0.25$$

at $x = 1$

$$y = 1^{-1} = \frac{1}{1} = 1$$

Average slope =

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{1 - 0.25}{1 - 0.5} = \frac{0.75}{0.5} = 1.5$$

Instant slope:

$$\begin{aligned} \frac{d}{dx} x^{-1} &= -1x^{-2} = \frac{-1}{x^2} \\ &\text{at } x = \frac{1}{2} \\ \text{the slope is } &\frac{-1}{\left(\frac{1}{2}\right)^2} = \frac{-1}{0.25} = -4 \end{aligned}$$

Q6 Suppose the graph of $y(x)$ climbs up to its maximum at $x = 1$. Then it goes downward for $x > 1$.

6A. What is the sign of $\frac{dy}{dx}$ for $x < 1$ and then for $x > 1$?

6B. What is the instant slope at $x = 1$?

before $x=1$, that is when $x<1$, y increases, the slope is positive. at $x=1$, the slope is 0. and when the $x>1$, slope is negative.

7 If $y = \sin x$, write an expression for $\frac{\Delta y}{\Delta x}$ at any point x .

We see later that this $\frac{\Delta y}{\Delta x}$ approaches $\cos x$

Practice problems from Prof. Gilbert Strang's Lecture - Max and Min